

Preventing Cross Network Slice Disruptions in a Zero-Trust and Multi-Tenant Future 5G Networks

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Abstract—As network slicing is the chief enabler for future Beyond 5G (B5G) and 6G networks, multiple tenants inter-operate cost-effectively to provide a variety of slice services on a common physical infrastructure. However, this opens the doors to cross-slice disruptions with Man-in-the-Middle (MITM) attack which ultimately disrupts the slice services in the data plane. In this paper, we address such possible cross-network slice disruptions in a zero-trust and multi-tenant based 5G network by proposing different design techniques namely, secure communication and Artificial Intelligence (AI)-based anomaly detection to prevent them. Our experiments on a 5G testbed prototype show that in the secure communication method, Attribute-Based Encryption (ABE) provides higher security benefits in confidentiality and implicit authorization. However, symmetric encryption and integrity protection prevent cross-slice disruptions with less communication overhead, but with a weaker security level. On the other hand, with online learning and noise tolerance capabilities, AI-based Hierarchical Temporal Memory (HTM) can proactively detect the occurrences of the identified cross-slice disruptions.

I. INTRODUCTION

Based on the variety of service requirements, multiple slices, and slice subnets are being deployed in 5G networks in an end-to-end slice setup including Radio Access Network (RAN) and 5G Core (5GC) portions. In this context, multiple slice tenants (or the slice owners) inter-op [1] to set up the end-to-end slicing. This kind of interoperability is highly encouraged when a single tenant may not want to or cannot reproduce building a fresh slice for the unprocurable or unavailable functions of a portion of an end-to-end slice. For example, as shown in Fig. 1, a set of tenants bring in the RAN side slice functions, a tenant provides the 5GC side slice in the control plane and a few more for the data plane. Here, the tenant from 5GC providing control plane services, may not have support for building the 5GC data plane and the RAN side functions. Hence, these kinds of multi-tenant based slice deployments are highly cost-effective for every tenant in providing end-to-end slice services. However, these interoperability-based deployments open the doors to security threats like Man-in-The-Middle (MITM) and side-channel attacks at cross-slice boundaries.

In this context, we aim to operate a secure end-to-end slice, where every participating slice tenant from different domains (RAN, 5GC) shall either have pro-active mechanisms to detect the possible anomalies occurring at the cross-slice boundaries or shall ensure to have secure inter-slice communication always. Specifically, we first identify the possible vulnerabilities

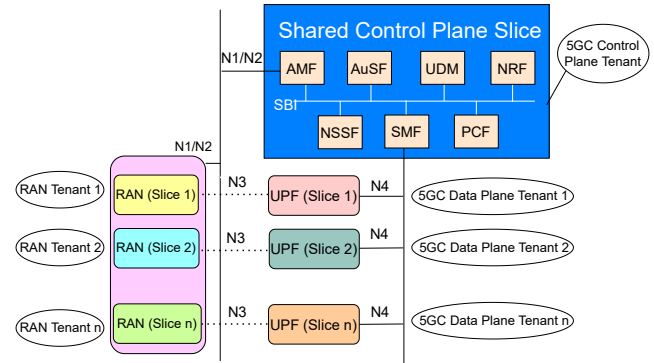


Fig. 1: Multi-tenant based 5G network slices interoperability.

that can occur at the cross-slice frontiers and then propose effective solutions to prevent them. Unlike traditional reactive approaches, the salience of all the proposed solutions is that they are proactive on the control plane communication itself, to prevent possible attacks on the data plane. Therefore, to summarize, the key contributions of this work are:

- Study of cross-slice disruptions in RAN and 5GC.
- Propose the following proactive techniques to prevent the identified cross-slice disruptions.
 - Symmetric Encryption and Attribute-Based Encryption (ABE) methods for data confidentiality between the slices.
 - Symmetric Integrity Protection for data integrity between the slices.
 - Anomaly detection using Artificial Intelligence (AI)-based Hierarchical Temporal Memory (HTM) [3].
- Evaluation of the proposed techniques on a 5G testbed prototype.

II. CROSS-SLICE DISRUPTIONS WITH MAN-IN-THE-MIDDLE

In this work, we specifically focus on the slice setup shown in Fig. 1. It can be noticed that the control plane functionality of 5GC is shared among all dependent slices on the data plane. As per 3GPP [4] Session Management Function (SMF) of 5GC (see Fig. 1) is responsible to establish various Protocol Data Unit (PDU) sessions for the User Equipment (UE) to access the respective slice services, with corresponding User Plane Function (UPF). Here, 3GPP [2] specifies that, during the respective PDU session establishment on the control plane,

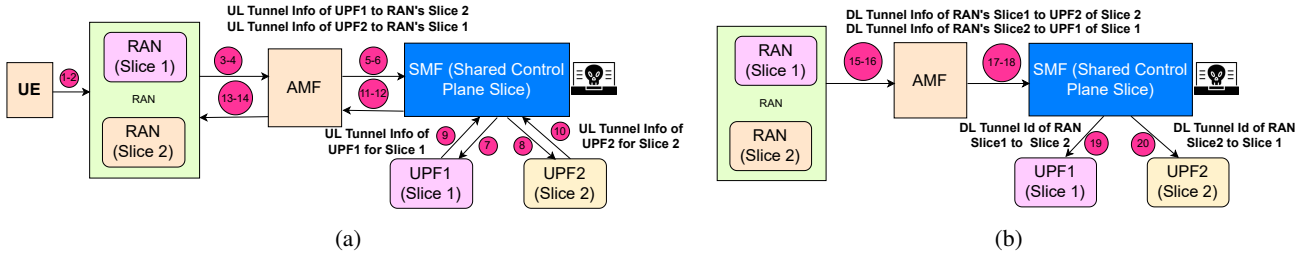


Fig. 2: Compromised SMF causing the cross-slice disruptions during PDU session establishment [2], to core slices (uplink) in a) and RAN slices (downlink) in b).

RAN and UPF of a matching slice have to mutually exchange their established tunnel data with each other via the SMF. Note that, tunnel data on either side (RAN and UPF) mainly consists of the tunnel ID and IP address of the respective network entity (RAN or UPF). This is done to allow the ultimate uplink and downlink data transfer on the data plane via the respective N3 interface. To this extent, SMF is fully aware of the N3 tunnel data exchanged between different UPFs and RANs (see [4, 2]) for the corresponding slices. A threat in this context arises when the SMF is not owned by the same slice tenant who owns the corresponding UPF and such a common SMF gets compromised. We now detail, how a compromised SMF can misuse the tunnel data exchanged between UPF and RAN.

- **Cross-Slice Access:** A compromised SMF can simply swap the tunnel data exchanged between UPFs and RANs across two different slices during the PDU session establishment procedures for these slices. See Fig. 2a where SMF is swapping the uplink tunnel data from the UPF side slice subnets towards the RAN side slice subnets after the 10th step. Fig. 2b shows the same for downlink tunnel data coming from RAN towards core side slice subnets represented by UPFs after the 18th step. As a result, in the data plane, the uplink and downlink data get exchanged between the slice subnets of different (or mismatched) types on the RAN and the core (UPFs). Such a data flow exchanged via mismatched slice subnets can result in Service Level Agreement (SLA) violations, without their own knowledge. Hence, there is a need for integrity protection of the tunnel data exchanged between RAN and UPF of a matching slice via a common SMF.
- **Flooding RAN and UPF:** As mentioned in [5] flooding in a mobile network is mainly in the form of Denial-of-Service (DoS) attacks which drain resources in either direction (RAN or UPF), ultimately disrupting the service. A compromised SMF can access the downlink tunnel data arriving from RAN during the PDU session establishment and share it with a UPF from an attacker. Using this tunnel data, the attacker can flood the RAN on the data plane (N3 interface) from the attacker's UPF. So, increased traffic on N3 beyond the capacity of RAN or UPF will disrupt the slice service ultimately. Eventually,

this will even stop legitimate users to connect to the respective slice, making it difficult to provide its 5G services. Similar flooding can be done on UPF in the uplink direction. Hence, there is a need for confidentiality of the tunnel data exchanged between RAN and UPF during the control plane PDU session establishment procedure.

III. RELATED WORK

While some 5G security challenges are identified in [5], authors in [6] identify a novel attack named Distributed Slice Mobility (DSM) attack by exploiting the UE-initiated inter-slice mobility. However, authors in [7] signify the need for a cross-slice authentication scheme at the edge cloud. While all these works focus on securing slice operations from unauthorized access, [8] identifies the interface between services using multiple slices as one of the key vulnerable points of attacks. Moreover, GSMA [9] encourages strong slice isolation with respect to security where the attack on a slice should not impact any other slice. Additionally, **zero-trust architecture** [10] is gaining great attention for building Cognitive Autonomous Networks (CANs) with cognitive functions [1]. Mapping this to a multi-tenant slice interoperability environment, it is important that every slice owner shall adopt the safety measures to keep the slice services secure and isolated. With these related works and motivation detailed in Section II, we dive deep into investigating the various cross-slice disruptions in the context of end-to-end slice deployment with RAN and 5GC in a multi-tenant based zero-trust interoperability environment.

IV. PROPOSED TECHNIQUES FOR PREVENTING CROSS-SLICE DISRUPTIONS

We now propose different proactive methods to tackle the cross-slice disruptions discussed in Section II.

A. Secure control plane communication between RAN and UPF

In this proposed method, RAN and UPF of a matching slice(s) securely communicate with each other on the control plane via the common SMF and AMF by encrypting and integrity protecting the corresponding data. First, a setup phase is executed, where the common Network Slice Management Function (NSMF), configures a common secret parameter,

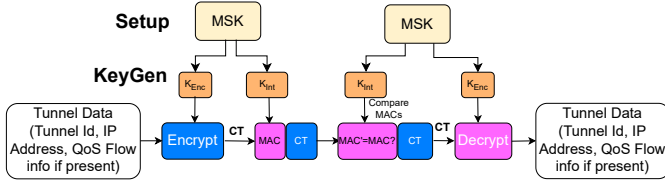


Fig. 3: Symmetric encryption and integrity protection.

Master Secret Key (MSK) at the RAN and UPF of the corresponding slice.

1) *Integrity Protection for resistance to cross-slice access:* When a new PDU session establishment request is received at UPF, it computes the Advanced Encryption Standard (AES) - Cipher-based Message Authentication Code (CMAC) of 16 bytes, for its entire tunnel data using 128-bit symmetric integrity protection key (K_{Int}). Fig. 3 shows the working of symmetric integrity protection in this context. Next, the UPF places the computed MAC and sends it to the SMF. This information is then conveyed from SMF to RAN through AMF in the PDU session resource setup request message using Next Generation Application Protocol (NGAP). On receiving this information, RAN in the proposed method computes the MAC locally, i.e., MAC' , and compares it with the received MAC from UPF, i.e., MAC . If it matches (i.e., if $MAC' = MAC$), then RAN is sure that the tunnel data is not altered in between. If it doesn't match, it rejects the PDU session resource setup request message. A similar integrity check is done at UPF for the downlink tunnel data coming from RAN. Suppose a compromised SMF swaps the tunnel information from the UPF of one slice with that of another slice and accordingly sends it to the RAN, the integrity check on the received tunnel at the RAN would fail. Hence the proposed method is resistant to cross-slice disruptions described in Section II.

2) *Confidentiality for resistance to flooding RAN and UPF on the data plane:* For the confidentiality of the tunnel data exchanged between RAN and UPF of corresponding slices, we propose two different encryption methods.

1) **Symmetric Encryption:** The UPF encrypts the tunnel data using the 128-bit Symmetric Encryption Key (K_{Enc}) and AES_CTR encryption algorithm. Note that K_{Enc} is generated from the secret parameter MSK configured during the setup phase. The advantage of this symmetric encryption method is that the length of the output data, i.e., encrypted data size is the same as the size of the input (plain) tunnel data. Hence, instead of actual tunnel data, UPF now sends the encrypted data for them, to the SMF. A similar flow is proposed from RAN to UPF to keep the downlink tunnel data confidential. Fig. 3 shows the working of symmetric encryption in this context. Suppose an attacker storing this encrypted tunnel data from compromised SMF uses it to flood the RAN, it cannot succeed, as the tunnel data is not the actual but the encrypted one and RAN would simply drop it. Hence the proposed method is resistant to flooding attacks described in Section II.

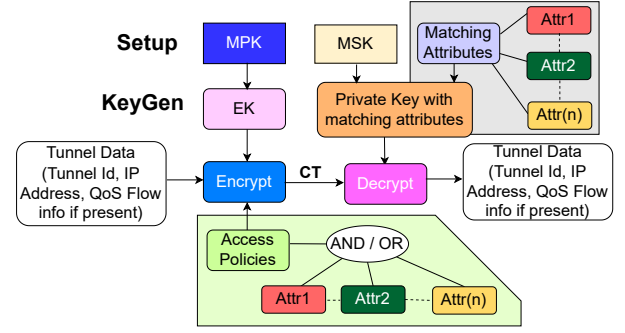


Fig. 4: Working of ABE with access policies and attributes.

2) **Attribute-Based Encryption (ABE):** ABE [11] is a flexible Public Key Encryption (PKE) method associated with attributes and access policies, providing multiple perks in the cryptography world, unlike the traditional PKE. ABE is highly feasible and quite encouraging to provide confidentiality of the tunnel data between RAN and UPF of the corresponding slice. Further, it can allow multiple UPFs acting as PDU session anchors and uplink classifiers [4] to decrypt the tunnel data if needed, without the need for multiple encryptions, unlike traditional PKE and symmetric approaches. We leverage Ciphertext-Policy ABE (CP-ABE), as it provides greater control to the data producer in setting the access policies. Fig. 4 shows the working of ABE with an access policy tree (can be a multilevel tree) having concatenation of AND and OR operators and attributes. The attribute set for RAN and UPF are defined as

$$RAN_{Attrset} = \{RAN_{ID}, NSSAI\}$$

$$UPF_{Attrset} = \{UPF_{IP}, UPF_{TunID}, NSSAI\}$$

In the working of ABE, first, the master public key MPK is generated and then the private key of the receiver is generated based on the MSK and the attributes possessed by the respective receiver. Using the public parameters and the access policy, UPF encrypts the tunnel data and generates the cipher text (CT). At the receiver (i.e., RAN) the decryption key (the private one embedded with respective RAN attributes) is used to decrypt the CT to produce the actual tunnel data sent from UPF. The decryption is successful only if RAN's attributes match with those specified in the access policy by the UPF during encryption. A similar flow exists in the RAN to UPF direction, where, UPF can decrypt the tunnel data sent by the RAN of the matching slice, only if UPF satisfies the access policy specified by RAN (during encryption) with respective UPF's attributes. Hence, along with confidentiality, the ABE is also very effective to provide implicit authorization that data is in fact sent by the corresponding slice entity only from the 5GC side.

B. HTM Learning for Cross-Slice Anomaly Detection

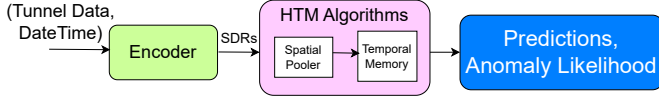


Fig. 5: Working of HTM for cross-slice anomaly detection.

Proposed always-on secure communication described in Section IV-A may add to control plane latency in mission-critical and delay-sensitive slice services like uRLLC and V2X. Therefore, we now describe an AI-based cross-slice anomaly detection method that overcomes the additional control plane latency but still can be effective by warning instantly whenever cross-slice anomalies occur.

Proposed AI-based HTM [3] is an unsupervised online learning method, that learns continuously to identify temporal anomalies on a sequence of real-time data like the arrival of 5G user requests in this work. HTM learning engine runs as part of the respective slice (i.e., both at RAN and UPF sides). For brevity, we pick only UPF. So, UPF sends the received tunnel data from RAN for HTM learning. Fig. 5 shows the working of HTM. As shown here, for HTM learning, *DateTime*, and tunnel data received at a time period t are picked as inputs. These input data are first encoded into Sparse Distributed Representation (SDR)s and then concatenated into a single SDR forming the concatenated input x_t which is then passed through a spatial pooler of HTM. The resulting vector is fed to the temporal memory component. This component models the temporal patterns to continuously learn the possible occurrence of anomalies.

V. PERFORMANCE EVALUATION

A. End-to-End Experimental Setup

To evaluate the performance of the proposed techniques, we developed an experimental setup on the in-house 5GC prototype testbed. This prototype is presented in our previous work [12] along with a RAN+UE emulator.

Symmetric Integrity Protection and Encryption: We relied on the OpenSSL APIs [13] to develop the symmetric integrity protection and encryption methods at UPF and RAN (see Fig. 3). For integrity protection, to communicate the computed MAC from UPF to RAN, we leverage the PFCP vendor-specific Information Element (IE) from UPF to SMF. Further, we added **Source to Target Transparent container IE** in the NGAP message carrying the UPF's tunnel data from SMF to RAN via AMF, as this change is required to implement the proposed solution. This IE contains the bytes of MAC along with its length so that the RAN can decode it appropriately. A similar implementation is done to carry the MAC of RAN's tunnel data to AMF. For symmetric encryption, the existing IEs of PFCP and NGAP were sufficient, as the length of encrypted data is the same as that of actual tunnel data.

Attribute Based Encryption (ABE): Here, we picked *cpabe* from [14], a toolkit providing four command line tools used to perform the various operations of the CP-ABE phases,

based on the native proposal of ABE [15]. We develop the invocable APIs on top of *cpabe* and then hook these APIs individually at RAN and UPF as per the flow shown in Fig. 4. Unlike the symmetric encryption method, the output of encrypted tunnel data from ABE is not the same as the length of the input tunnel data. Hence, here too we make use of transparent container IE(s) proposed in symmetric integrity protection Section V-A, to carry this ABE encrypted data between UPF and RAN.

The entire test framework in this work is set up as shown in Fig. 1. On the 5GC, we deploy two slices in the data plane and a common control plane slice. Every slice is hosted on Intel® Xeon® CPU E5-1650 v4 @ 3.60GHz with 6 CPU cores and 32GB RAM. All the NFs of 5GC and the RAN+UE emulator run as a docker container. Note that, we did not deploy slices on RAN as the slices on the 5GC (with UPFs and control plane) are sufficient to show the cross-slice disruptions and evaluate the proposed techniques.

B. Performance of Secure Communication

For evaluation, we measure the session establishment time and the communication overhead incurred during the PDU session establishment as this phase in the control plane sets up the ultimate data plane path for slice service.

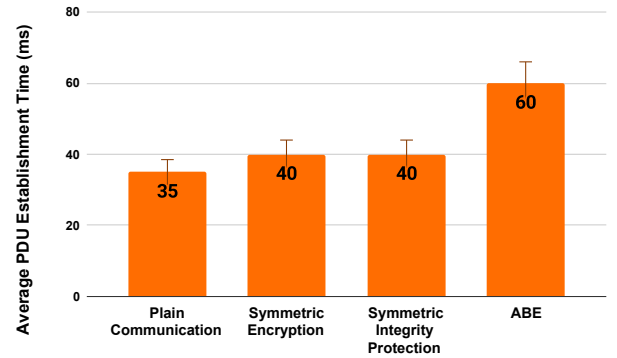


Fig. 6: PDU session establishment time by various methods.

1) **PDU Session Establishment Time:** PDU session establishment time is the total time it takes for a single PDU session to be established between UE and the 5GC in the end-to-end procedure [2]. Note that radio side time duration is not considered here as the RAN+UE emulator does not have the radio stack. Fig. 6 compares the average PDU session establishment time between various methods. It confirms that the proposed secure communication methods do not actually add to this computation overhead except slightly high in ABE.

2) **Communication Overhead:** Communication overhead depicts the number of bytes that a method occupies in exchanging the tunnel data between RAN and UPF via the control plane. Table I compares this metric between various methods proposed against the plain one (without the proposed ones). It can be observed that ABE occupies the highest number of bytes unlike symmetric ones as ABE employs bi-linear pairing operations which are computationally intensive.

Overall, we realize that ABE produces minimal cipher text when the length of the input access policy (with applicable attributes) and input data are quite minimal.

TABLE I: Comparison of communication overhead (in bytes)

Direction	Plain	Symmetric Encryption (AES) [13]	Symmetric Integrity Protection (AES) [13]	ABE (cpabe [14])
UPF to RAN	8	8	24	858
RAN to UPF	10	10	26	1157

C. Performance of HTM in Anomaly Detection

We implemented the HTM model using HTM Core [16]. It is a C++ library with Python bindings. To evaluate the performance of HTM, we had to generate the data set. We consider the RAN's IP address of the N3 tunnel and the respective date and time of its arrival. To simulate the realistic cross-slice disruptions happening at different hours of the day, when different slice types (eMBB, uRLLC, mMTC) operate in parallel, we set the 24 hours time pattern with hourly units up to 5 months. Note that, here time units can be set in other units too like minutes. As per the referred works in [17], the usage of mMTC slice is expected to be very less at night. Introducing cross-slice anomaly here, we increase the use of mMTC slice once a week during the night hours. This is essentially the situation described in Section II where the compromised SMF takes the benefit of one slice (mMTC) over the other (like uRLLC).

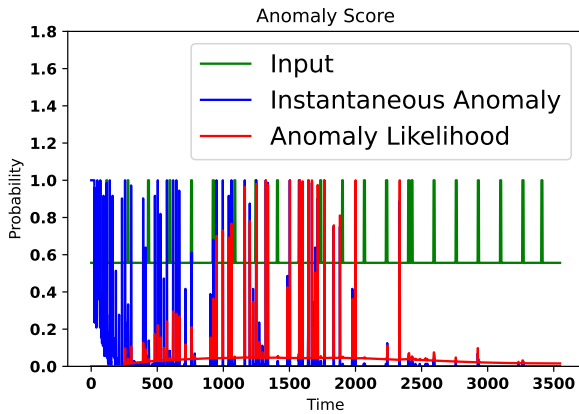


Fig. 7: IP address change anomaly detection by HTM.

We set the learning period in HTM to 250 time units. HTM outputs the probability of instantaneous anomaly and anomaly likelihood in the slice. The closer these numbers are to 1, the higher the chance it is an anomaly. As shown in Fig. 7, during the early learning period, HTM learning predicts more instantaneous anomalies. But the probabilities of anomaly likelihood are higher and very close to actual anomalies after the learning period. Such, consistent higher values of anomaly likelihood at the early stages are very useful in detecting anomalies.

VI. CONCLUSION AND FUTURE WORK

In this paper, we identified a cross-slice security threat between RAN and UPF of 5GC caused by the compromised SMF and proposed effective solutions to prevent them. Experiments on the 5G testbed prototype prove that, in secure control plane communication, compared to the symmetric method, ABE provides higher security benefits in confidentiality and implicit authorization, but with additional communication overhead. So, ABE can be leveraged in inter-slice communication for smaller-length data, without communication constraints. On the other hand, with continuous learning on the arriving user requests, AI-based HTM is a highly promising approach in detecting cross-slice anomalies. In the future, we plan to extend our proposed solutions for possible cross-slice disruptions in the inter-slice handover scenarios.

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